FIFA World Cup Analysis

Insights & Data Model Report

Based on the Power BI file **Fifa_world_cup_completed_v2.pbix**: 21 men's World Cups from 1930 to 2018, with match, goal, card, team, player and stadium data. All figures below were recomputed directly from the tables inside the model.

21 Tournaments	**900** Matches	**2,548** Goals	**2.83** Goals / match
84 Teams	**7,907** Players	**185** Stadiums	**2,466** Cards (1970+)

Key takeaways

1. **Scoring peaked early, then settled.** Matches averaged 5.38 goals in 1954 but only 2.21 in 1990. Since then every World Cup has landed between 2.2 and 2.7, against an all-time average of 2.83.
2. **Brazil is the benchmark.** Five titles, 76 wins in 109 matches (69.7%), 229 goals, and the only team present at all 21 tournaments. Germany combined (West Germany 3 + Germany 1) and Italy each have 4 titles.
3. **Hosting is a real advantage.** Host teams won 67% of their matches versus 39% for everyone else (+28 points), and 6 of 22 host entries went on to win the cup.
4. **Goals arrive late.** The 76-90 minute window produces 21% of all goals versus 13% in the opening 15 minutes. 604 goals (23.7%) came in the 76th minute or later, including extra time.
5. **Knockout football is higher-stakes and higher-scoring.** Semi-finals, third-place matches and finals average 3.5-3.8 goals, against 2.69 in the group stage. 68 matches went to extra time and 30 to penalty shootouts.
6. **Discipline escalated after 1970.** Cards per match climbed from 1.6 (1970) to a peak of 5.5 in 2006. In total there were 148 dismissals: 89 straight reds and 59 second yellows.

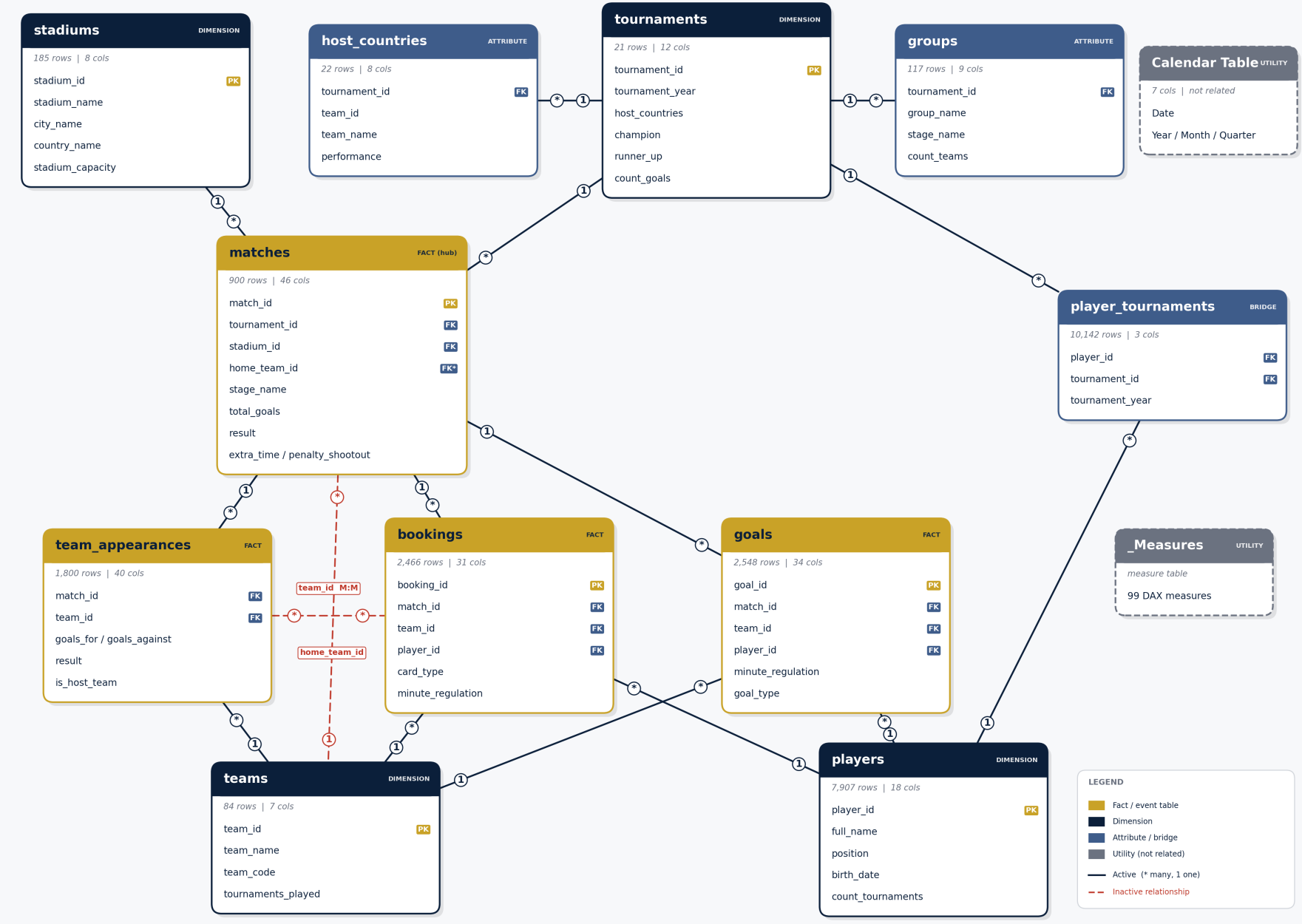
Roll of honour

Year	Host	Champion	Runner-up
1930	Uruguay	**Uruguay**	Argentina
1934	Italy	**Italy**	Czechoslovakia
1938	France	**Italy**	Hungary
1950	Brazil	**Uruguay**	Brazil
1954	Switzerland	**West Germany**	Hungary
1958	Sweden	**Brazil**	Sweden
1962	Chile	**Brazil**	Czechoslovakia
1966	England	**England**	West Germany
1970	Mexico	**Brazil**	Italy
1974	West Germany	**West Germany**	Netherlands
1978	Argentina	**Argentina**	Netherlands
1982	Spain	**Italy**	West Germany
1986	Mexico	**Argentina**	West Germany
1990	Italy	**West Germany**	Argentina
1994	United States	**Brazil**	Italy
1998	France	**France**	Brazil
2002	Japan & South Korea	**Brazil**	Germany
2006	Germany	**Italy**	France
2010	South Africa	**Spain**	Netherlands
2014	Brazil	**Germany**	Argentina
2018	Russia	**France**	Croatia

What is inside this report: (1) the data model diagram and a table-by-table breakdown, (2) nine insight areas with charts, and (3) a model review with concrete fixes for issues found in the DAX and relationships.

FIFA World Cup Analysis - Power BI Data Model

11 data tables | 1 calendar | 1 measures table | 16 relationships (14 active, 2 inactive) | 99 DAX measures



1. Data model at a glance

The model is a **snowflake-style schema** with **matches** as the hub. Three event tables (goals, bookings, team_appearances) hang off matches; tournaments, stadiums, teams and players act as dimensions. All active relationships are many-to-one with single-direction filtering, which is the healthy default.

Table	Role	Rows	Cols	Grain (one row per...)
matches	Fact (hub)	900	46	match
team_appearances	Fact	1,800	40	team per match (2 rows per match)
goals	Fact	2,548	34	goal (incl. 52 own goals)
bookings	Fact	2,466	31	card shown (1970 onward)
player_tournaments	Bridge	10,142	3	player per tournament squad
players	Dimension	7,907	18	player
teams	Dimension	84	7	national team
tournaments	Dimension	21	12	World Cup edition
stadiums	Dimension	185	8	stadium
host_countries	Attribute	22	8	host nation per tournament
groups	Attribute	117	9	group per tournament
Calendar Table	Utility	-	7	date (not related to any table)
_Measures	Utility	-	1	holds the 99 DAX measures

Relationships (16)

From (many)	To (one)	Key column	Status
matches	tournaments	tournament_id	Active
matches	stadiums	stadium_id	Active
team_appearances	matches	match_id	Active
goals	matches	match_id	Active
bookings	matches	match_id	Active
team_appearances	teams	team_id	Active
goals	teams	team_id	Active
bookings	teams	team_id	Active
goals	players	player_id	Active
bookings	players	player_id	Active
player_tournaments	players	player_id	Active
player_tournaments	tournaments	tournament_id	Active
host_countries	tournaments	tournament_id	Active
groups	tournaments	tournament_id	Active
matches	teams	home_team_id to team_id	**Inactive**
team_appearances	bookings	team_id (many-to-many, both directions)	**Inactive**

Inactive relationships are switched on only inside measures that call USERELATIONSHIP.

Report and measures

8 report pages, 125 visual objects (including shapes and text boxes):

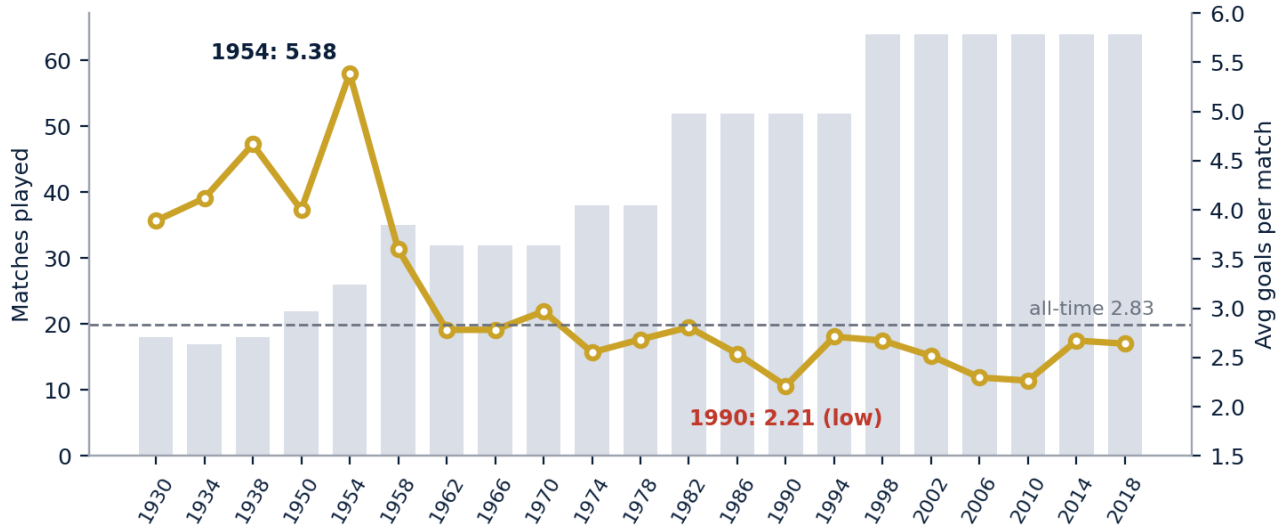
- Cover (9)
- Teams Across Tournaments (17)
- Stadiums & Geography (16)
- Host Countries & Teams (15)
- Groups & Match Results (17)
- Goals & Timing (17)
- Players & Discipline (17)
- Advanced Insights (17)

99 DAX measures in _Measures, by theme:

- Teams, hosts & rankings: 28
- Matches & tournaments: 21
- Goals: 15
- Discipline (cards): 9
- Players: 7
- Half-split goals: 6
- Trends vs previous edition: 5
- Home / away splits: 5
- Selectors: 2 | Red Cards: 1

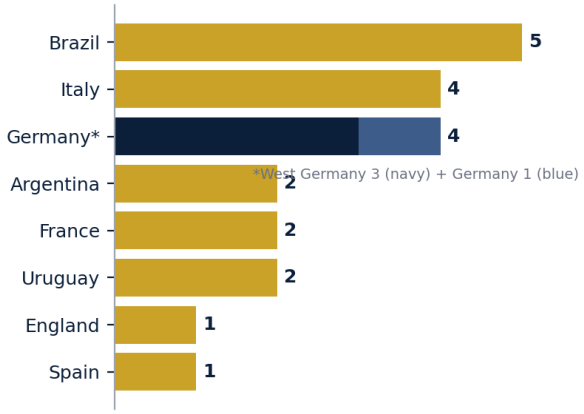
2. Insights

2.1 Scoring: a golden age, then a plateau



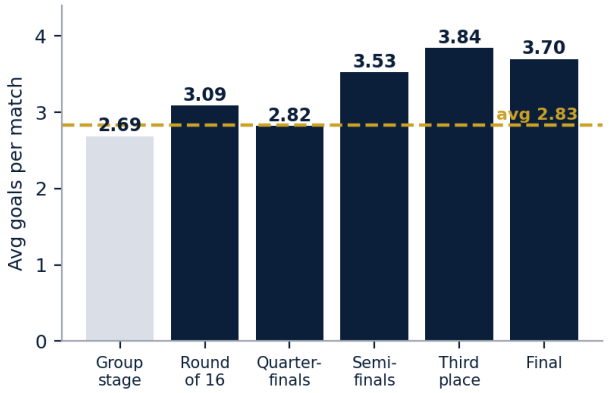
The first four tournaments averaged over four goals per match; 1954 (Switzerland) remains the record at 5.38 with 140 goals in only 26 matches. The format then grew from 16 to 24 teams (1982) and 32 teams (1998) while the scoring rate fell and flattened. The most recent edition, 2018, averaged 2.64.

2.2 Champions



Eight nations have won the cup and only Brazil has won more than four times. Counting West Germany and Germany as one team puts Germany level with Italy on four.

2.3 Knockout rounds score more



Every knockout round beats the group-stage average. The third-place match is the highest scoring at 3.84 goals.

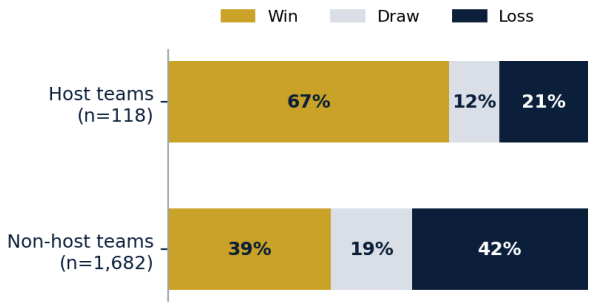
2.4 All-time team records (ranked by wins)

Team	Titles	Played	W	D	L	GF	GA	Win %
Brazil	5	109	76	14	19	229	105	69.7%
Argentina	2	81	47	10	24	137	93	58.0%
Italy	4	83	46	17	20	128	77	55.4%
West Germany	3	62	39	11	12	131	77	62.9%
France	2	66	36	9	21	120	77	54.5%
Germany	1	47	32	5	10	95	48	68.1%
Spain	1	63	31	11	21	99	72	49.2%
England	1	69	30	17	22	91	64	43.5%
Netherlands	0	50	28	9	13	86	48	56.0%

Team	Titles	Played	W	D	L	GF	GA	Win %
Uruguay	2	56	25	11	20	87	74	44.6%

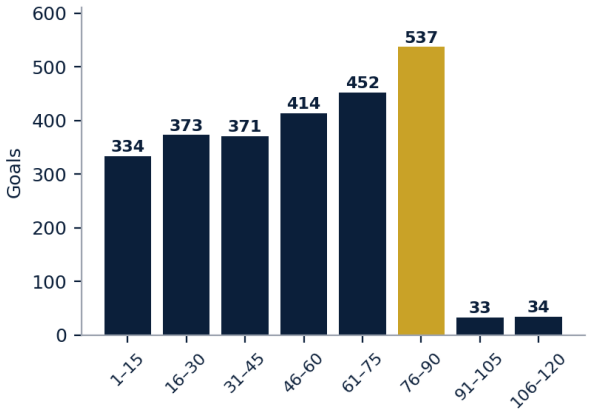
Note: West Germany (1954-1990) and Germany are separate rows in the model. Combined they have 71 wins from 109 matches and 226 goals, which is close to Brazil's 76 wins and 229 goals.

2.5 Host advantage



Hosts win 67% of matches against 39% for non-hosts. Italy won 10 of 12 home matches and France 8 of 9.

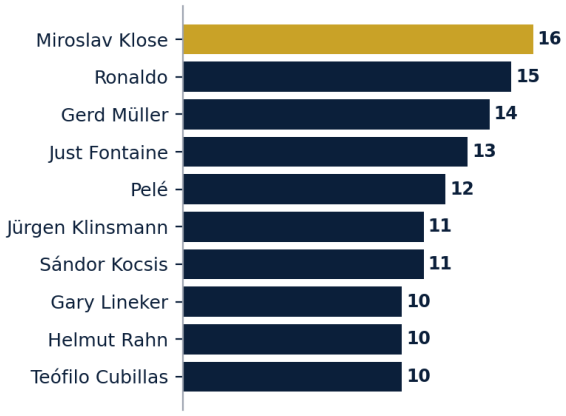
2.6 When goals are scored



Scoring rises through the match: 334 goals in minutes 1-15 versus 537 in 76-90, a 61% increase. Fatigue and urgency both show up.

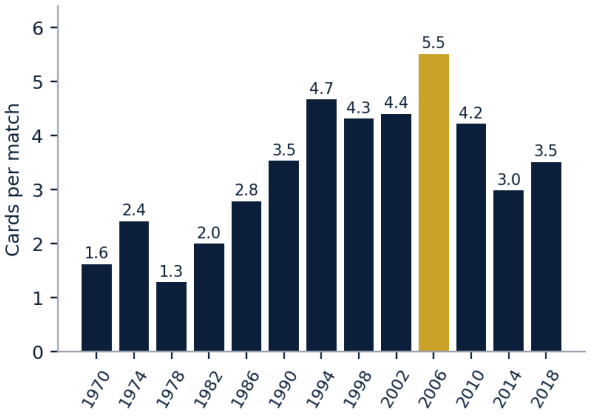
Goal types: open play 2,299 (90.2%), penalties 197 (7.7%), own goals 52 (2.0%). **By period:** first half 1,078 (42.3%), second half 1,403 (55.1%), extra time 67 (2.6%). **Shootouts:** 30 matches; Argentina won 4 of its 5.

2.7 Top scorers (excluding own goals)



Miroslav Klose leads with 16 goals across four tournaments. Just Fontaine's 13 goals in 1958 is still the single-tournament record.

2.8 Discipline



Card data starts in 1970. 2006 was the peak at 5.5 cards per match; two matches reached 16 cards (Cameroon v Germany 2002, Portugal v Netherlands 2006).

2.9 Players, stadiums and scale

- **52 hat-tricks** (3+ goals by one player in a match), 7 of them with 4 or more.
- **7,907 players** have appeared; 6,138 (78%) played in just one World Cup and only 4 played in five.
- Biggest wins: Hungary 9-0 South Korea (1954), Yugoslavia 9-0 Zaire (1974), Hungary 10-1 El Salvador (1982).
- Highest scoring match: Austria 7-5 Switzerland (1954), 12 goals.
- **185 stadiums** in 160 cities across 17 countries; average capacity per match is 55,581.
- Largest venue: Maracana (200,000). Most matches hosted: Estadio Azteca (19), then Maracana (15).
- Germany has hosted the most matches (102, across 1974 and 2006), ahead of Brazil (86) and Mexico (84).
- 71 goalless matches (7.9%); 39.7% of games are decided by a single goal.

Reading note on "home team": at a World Cup most matches are played at neutral venues, so the home/away label is nominal. The 57% 'home team win' rate (513 of 900) partly reflects hosts and seeded teams being listed as home, so treat it as 'listed-first team' rather than a true home effect.

3. Model review and recommended fixes

The model is well structured and the report logic is mostly sound. These items were found by reading the relationships and DAX; they are ordered by how likely they are to change a number on the report.

Priority	Finding	Why it matters	Recommended fix
High	Goals first half / second half / extra measures (and their % versions) sum team_appearances[goals_for] filtered by bookings[match_period], using the inactive many-to-many team_id link.	bookings is the card table. The measure keeps any team that received a card in that period and sums its full goals_for, so it does not reflect when goals were actually scored.	Rebuild the three measures on the goals table (code below). Then the inactive M:M relationship is no longer needed and can be deleted.
Medium	Home Win % divides by [Matches Played Home] (uses USERELATIONSHIP to teams) while Away Win % and Draw % divide by [Matches Played] (no team link).	With a team slicer the home and away percentages can be based on different row sets and will not add up to 100%.	Use one denominator and one relationship pattern for all three. Consider renaming to 'Listed-home win %' (see the reading note).
Medium	Calendar Table has no relationship; the model also carries two auto date/time tables (LocalDateTable) and a DateTableTemplate.	Time intelligence cannot use Calendar Table, and the hidden auto tables add size and clutter.	Relate Calendar[Date] to matches[match_date], mark it as the date table, and turn off Auto date/time in Options.
Medium	West Germany, Germany, Soviet Union / Russia and Yugoslavia / Serbia are separate teams.	Titles, wins and top-team rankings split a single football lineage across rows.	Add a 'team_lineage' column in teams (for example West Germany and Germany to Germany) and use it in team visuals.
Low	groups is linked only to tournaments; host_countries[team_id] and matches[away_team_id] have no relationship.	Group tables and away-team views cannot be cross-filtered from teams or matches.	Add a composite key (tournament_id + group_name) for groups to matches, and relate host_countries[team_id] to teams. Add away_team_id as a second inactive link.
Low	Fact tables repeat descriptive columns (tournament_name, match_name, team_name, stadium details, stage and group).	Larger model, harder maintenance, and a risk of the same label being edited in one place only.	Keep keys and numeric fields in the facts and take names from the dimensions.
Low	Housekeeping: 'Red Cards' duplicates 'Straight Reds'; measure named 'Home Loass %'; all 99 measures have no display folder.	Cosmetic, but it makes the field list harder to use.	Delete the duplicate, fix the typo, and group measures into folders using the themes on page 3.

Corrected half-split measures

```
Goals first half  = CALCULATE ( [Goals Scored],
    goals[match_period] IN { "first half", "first half, stoppage time" } )
Goals second half = CALCULATE ( [Goals Scored],
    goals[match_period] IN { "second half", "second half, stoppage time" } )
Goals extra time  = CALCULATE ( [Goals Scored],
    goals[match_period] IN { "extra time, first half", "extra time, second half",
                             "extra time, second half, stoppage time" } )
Goals first half %  = DIVIDE ( [Goals first half],  [Goals Scored] )
Goals second half % = DIVIDE ( [Goals second half], [Goals Scored] )
Goals extra time %  = DIVIDE ( [Goals extra time],  [Goals Scored] )
```

Expected result from the data: first half 1,078 goals (42.3%), second half 1,403 (55.1%), extra time 67 (2.6%). These three add up to all 2,548 goals.

Data limits to keep in mind

- Cards are only recorded from 1970 (13 tournaments), so card measures already filter on year 1970+ for per-match rates. Keep that filter on any new card measure.
- The data stops at 2018 and covers the men's tournaments only; 2022 is not in the model.
- There is no attendance, club or minutes-played data, so questions about crowds or player workload cannot be answered from this model.
- Stadium capacity is a per-stadium figure, not match attendance.